Week 6 — In-Class Exercise

The funnel by hand · writing assumptions · which model

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Monday, October 19, 2026 · 20 minutes · groups of 3–4 · one worksheet per group

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All data are synthetic. LYMBREXA is a fictional third-in-class CD20×CD3 bispecific for 3L+ DLBCL, launching January 2027. Numbers are rounded; the point is the logic, not the decimals.

Part A · The funnel by hand (8 minutes)

Compute LYMBREXA's 2027 numbers with the same formulas as the model. Progression takes one year, so 2027's 2L and 3L+ starts come from 2026's patients.

Step	Formula	Inputs	Result
A1. 1L patients 2027	new cases × diagnosed × treated	22,910 × 97% × 86%	
A2. 2L starts 2027	1L patients 2026 × progress to 2L	18,690 × 31%	
A3. 3L+ starts 2027	2L starts 2026 × progress to 3L × later lines	5,670 × 50% × 1.24	
A4. New patients 2027	3L+ starts × peak share × year-1 uptake × coverage	A3 × 15% × 30% × 70%	
A5. Patients on therapy	new patients × months on therapy ÷ 12	A4 × 6 ÷ 12	
A6. Net revenue 2027	new patients × months × compliance × price × gross-to-net	A4 × 6 × 95% × $18,500 × 76%	

A7. Why is A5 smaller than A4? Which of the two does supply plan on, and which is the sales team paid on? One line.

A8. Which single assumption would you research first, and where would you look? One line.

Part B · Write three assumptions (7 minutes)

Fill one row each for peak share of 3L+ starts, year-1 uptake, and average months on therapy. Use the evidence below. Give low / base / high, the source type, and a one-line reason.

Evidence	What it says
Order of entry (Kalyanaram, Robinson & Urban 1995)	entrant share ≈ pioneer share ÷ √(order of entry). The first bispecific's peak share: 26%.
Analog launches: share of peak in year 1	outpatient bispecifics 24–37%; CAR-T 8–12% (63% of eligible patients never referred — IQVIA 2025)
Months on therapy (Kaplan–Meier, three analog regimens)	means 5.9, 8.3 and 9.4 months; LYMBREXA's label: fixed course of 8 monthly cycles

Assumption	Low	Base	High	Source type	Reason
Peak share of 3L+ starts					
Year-1 uptake (share of peak)					
Months on therapy					

Part C · Which model (5 minutes)

For each situation: incidence or prevalence; flow or stock; budget or long-range plan; and the simplest method that works.

Situation	Answer
1. A diabetes brand in year 4; no launches or patent expiry coming	
2. A gene therapy for an ultra-rare disease, ~400 US patients	
3. An oral brand 9 months from patent expiry	
4. Monthly targets for 2,000 accounts of an in-line brand	
5. A 3L+ oncology brand gaining a 2L label next year	
6. A competitor reads out Phase 3 against your brand next spring	